Capstone Two Final Project Report
U.S. Lower 48 Hourly Electricity Demand Forecasting

Executive Summary

This project develops a machine learning model to forecast hourly electricity demand across the U.S. Lower 48 states. Accurate electricity demand forecasting is critical for grid reliability, economic efficiency, and operational planning. Three regression models were developed and evaluated using time-series validation techniques: Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor. The Random Forest model achieved the best predictive performance with an R^2 of 0.944 and the lowest prediction error metrics, demonstrating strong capability for forecasting electricity demand.

Problem Statement

Electricity demand varies throughout the day and across seasons due to human activity patterns and energy production changes. Grid operators require reliable demand forecasts to optimize generation resources, reduce costs, and maintain system stability. The goal of this project is to build a predictive model that estimates hourly electricity demand using historical generation data and time-based features.

Dataset Description

The dataset contains hourly observations of electricity demand in megawatts along with generation by energy source including coal, gas, nuclear, hydro, solar, wind, petroleum, and other sources. Datetime information was used to engineer additional time-based features including hour of day, day of week, month, and a weekend indicator.

Modeling Approach

A time-series modeling approach was used to avoid data leakage. The dataset was split chronologically with the first 80% used for training and the remaining 20% used for testing. Three regression models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 metrics.

Final Model Selection

The Random Forest Regressor was selected as the final model due to its superior performance across evaluation metrics. It achieved the highest R^2 and lowest prediction errors compared with Linear Regression and Gradient Boosting models.

Conclusion

This project demonstrates that ensemble machine learning models can effectively forecast electricity demand. The Random Forest model captured nonlinear relationships between energy generation sources and time-based features. Future improvements may include incorporating weather data and additional lag-based time series features.

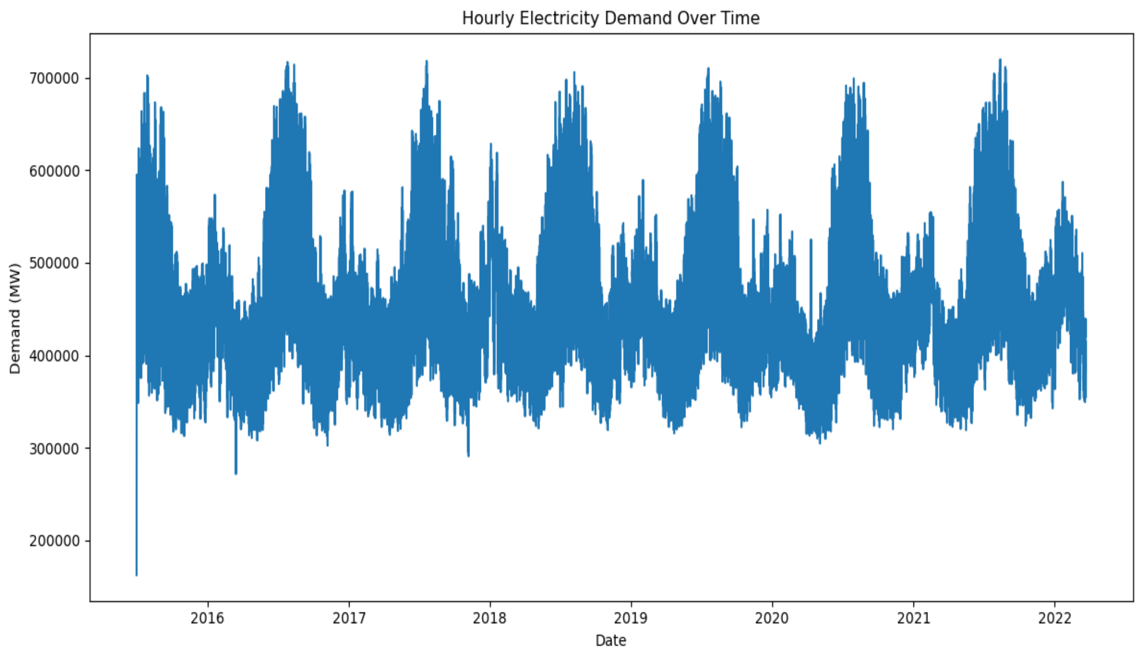

Figure 1: Hourly Electricity Demand Over Time

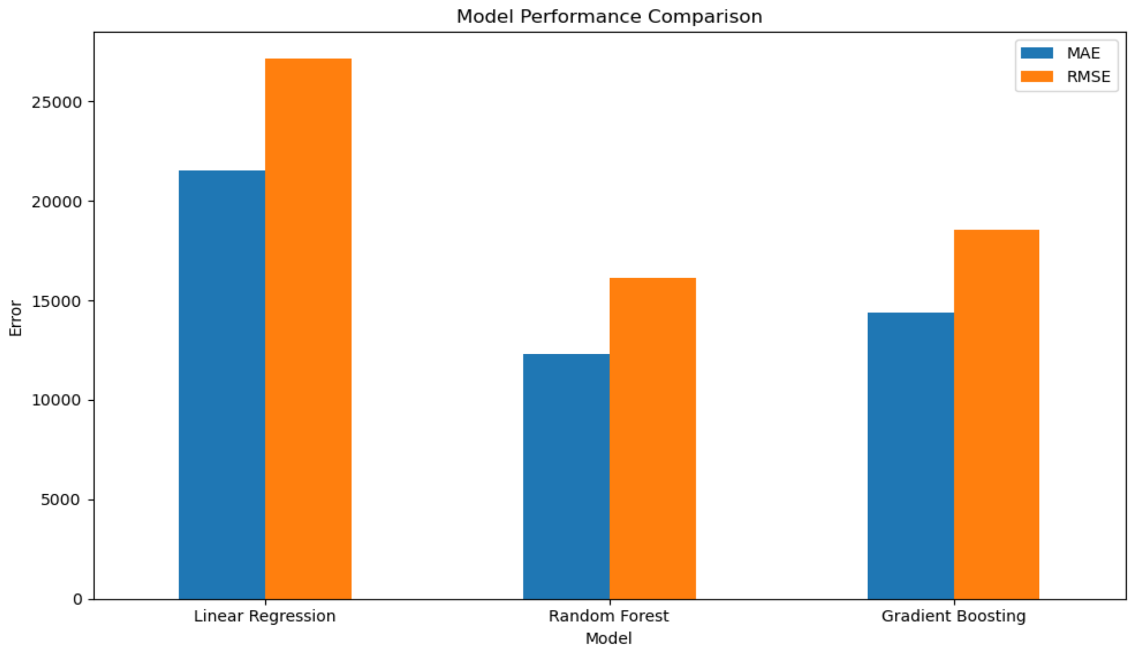

Figure 2: Model Performance Comparison

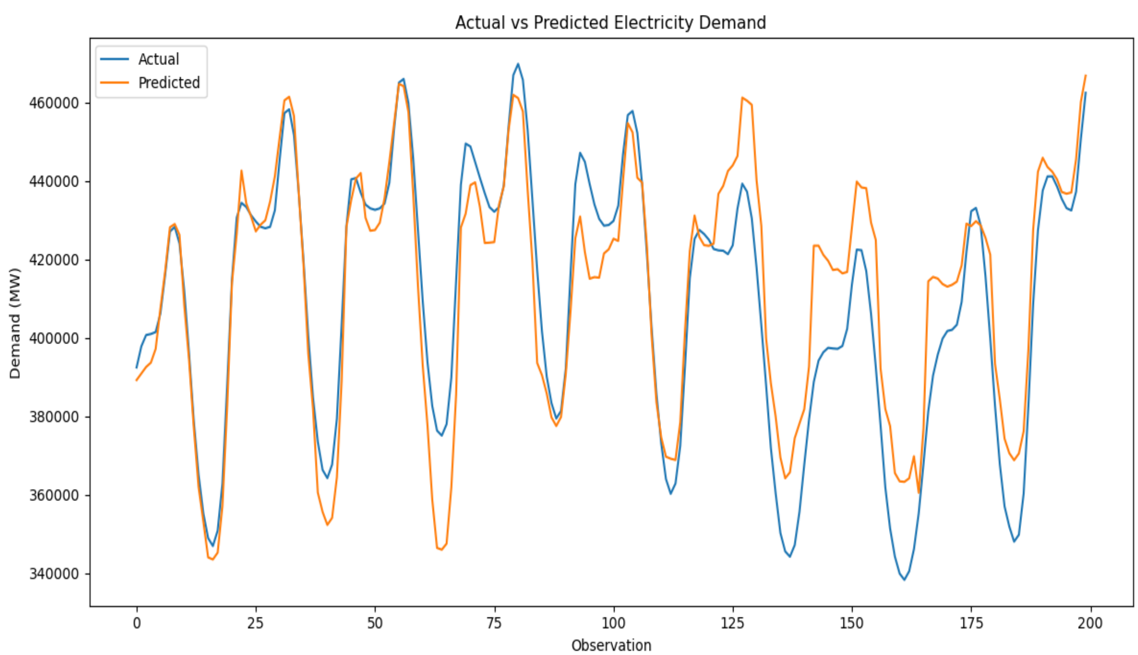

Figure 3: Actual vs Predicted Electricity Demand